



温州肯恩大学
WENZHOUCHEAN UNIVERSITY

Course Information

Course Title: Python Programming for Business

Course Number and Section: MGS 3001 WHS01

Campus Location: Wenzhou-Kean University

Semester: 2026 Spring

Meeting Days and Times: Tue & Thu 10:00 – 11:15

Meeting Location: CBPM A102

Instructor Information

Instructor Name: Dawei CHEN

Office Location: CBPM B214

Office Hours: Tue & Thu 11:15 – 12:30 | Wed 14:00 –16:30

Email: dchen@kean.edu

Course Description

Programming is a core component of the analytics field, as it facilitates data cleaning, preparation, and mining. Python scripting language is widely used in the data science field for data exploration and analysis. Through this course, students learn fundamental programming concepts and python programming syntax. The course familiarizes the students with Python's built-in modules, libraries, and functions that are essential for pattern and knowledge discovery.

Course Level Student Learning Outcomes

Upon successful completion of this course, the student will be able to:

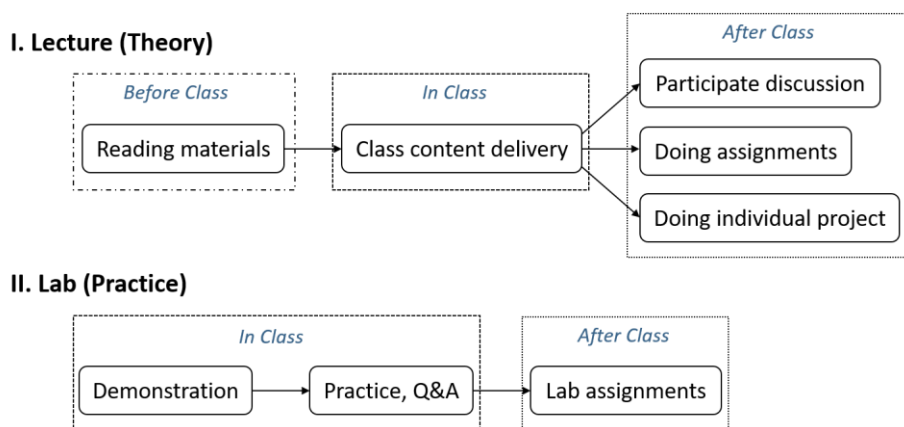
- Generate python scripts to extract, clean and prepare data
- Write code for processing data in spreadsheets and CSV files
- Use Python modules for parsing files and grouping data
- Analyze and summarize the data using tabular and graphical display
- Recall and use standard library modules and built-in functions

Section Level Student Learning Outcomes

Upon successful completion of **this Honor section**, students will be able to:

- Leverage **agentic (AI-assisted) coding** to design, generate, and evaluate Python scripts for data tasks, shifting from traditional syntax-focused programming to problem-oriented execution.
- Conduct **an end-to-end empirical business research project**, including formulating a research question, and using Python to collect, clean, analyze, and visualize data.
- Produce and present **a conference-style paper** that clearly communicates methods, findings, and implications.

Instructional Method



Course Materials

No book required. Reading materials will be provided throughout the semester as needed.

Course Schedule

Date		Topics	Note
w1	2/26	Course Introduction & House Rules	Icebreaker, course objectives, Honor section positioning, course assessment
w2	3/03	Introduction to Python	Environment setup (VS Code, Jupyter Notebook), first Python script, syntax
	3/05	Python Fundamentals	Variables, core data types, operators, comments
w3	3/10	Data Structures & Control Flow	Lists, tuples, sets, dictionaries; if/else, loops (for, while, break, continue, pass)
	3/12	Functions	Functions, parameters, arguments, return, lambda expressions, list comprehensions
Assignment 1 (research idea & initial proposal draft) is due on 3/15 (Sunday) at 23:59.			

w4	3/17	Research Workshop I	Research questions, literature review, contribution
	3/19	Object-Oriented Programming	Class, object, attributes, methods
w5	3/24	Debugging & Error Handling	Try/except, trackbacks, debugging strategies
	3/26	Files & Modules	Read/write text/CSV, file operations, modules
w6	3/31	AI Fluency	"AI Fluency: Framework & Foundations" by Anthropic
	4/02	Agentic AI Coding	Agentic AI coding workflow with VS Code, Jupyter Notebook and GitHub
w7	4/07	Midterm Review	
	4/09	Midterm Exam	Closed-book, in-class written exam (code reading, debugging and understanding)
	4/11	Midterm Feedback + AI Fluency cont.	
w8	4/14	Research Workshop II	Hypothesis, measurement
	4/16	Big Picture of Web Data Collection & GitHub Literacy	Navigate GitHub, find and evaluate repositories, fork/clone, run open-source notebooks in VS Code
Assignment 2 (revised proposal: data & empirical strategy) is due on 4/17 (Friday) at 23:59.			
w9	4/21	API-Based Data Collection	HTTP request/response, static vs. dynamic sites, authentication, GitHub API demo, pagination
	4/23	JSON & Semi-Structured Data	JSON structure, parsing nested data, flattening semi-structured data into tabular format, export to CSV
w10	4/28	WKU Student Research Day	No lecture
	4/30	HTML Scraping	Tags, attributes, CSS selectors, requests + BeautifulSoup
w11	5/05	<i>International Labor Day (No Class)</i>	
	5/07	Structured Data Cleaning	Pandas, DataFrame, missing values, duplicates, type conversion, outliers
Assignment 3 (data collection report) is due on 5/10 (Sunday) at 23:59.			
w12	5/12	Text Processing & Variable Construction	Regex, string methods, construct numeric variables from text
	5/14	Exploratory Data Analysis	Descriptive statistics, correlations, groupby analysis
w13	5/19	Data Visualization	Matplotlib, Seaborn
	5/21	Regression: Theory & Interpretation	Coefficients, p-values, R^2 , model specification
w14	5/26	Regression: Application	statsmodels OLS on project data, interpret results
	5/28	Research Workshop III	Academic paper anatomy, section-by-section guidance, writing with AI
w15	6/02	Research Workshop IV	Draft feedback, presentation prep, consultation
	6/04	Final Presentation (No Final Exam)	6-min presentation + 2-min Q&A per project
w16	6/09		

Final report is due on 6/10 (Wednesday) at 23:59.

**The schedule is tentative and may be adjusted based on class progress.*

Course Assessment

Grade Breakdown

Participation	10%
Assignments	3×10%
Midterm Exam	30%
Final Project (presentation + report)	30%
Total	100%

1. Participation (10%)

- Students are expected to attend all class sessions and actively participate.
- Leave requests must be submitted via email (dchen@kean.edu) with appropriate supporting documentation.
- 1 point will be deducted for each unapproved absence.
- 3 consecutive unapproved absences or missing an exam will be reported to the University in accordance with academic policy.

2. Assignments (30%)

- 3 assignments serve as milestones for the final project (10% each). Please refer to the final project guidelines for details.
- Late submissions will receive a 50% deduction.

3. Midterm Exam (30%)

- Tentative date: 4/09 (Thursday)
- Format: closed-book, in-class written exam to test understanding of Python codes.
- No make-up exam will be provided.

4. Final Project (30% = report 15% + presentation 15%)

This project is an individual empirical study examining the business, economic, or societal impact of artificial intelligence (AI). The final deliverables include a short conference-style paper (15–20 pages including everything, 12-point font, 1.5-spaced) and a brief presentation.

Throughout the semester, each student will complete the core components of an empirical research project. The process is iterative rather than strictly linear, and students are encouraged to refine earlier steps as the project develops. Three milestone assignments and three research

workshops are integrated into the course to provide structured guidance and ensure steady progress, with instructor support at each stage.

The project involves the following stages:

- Formulate a clear research question related to AI's impact;
- Develop a research proposal outlining the motivation, brief literature background, data sources, and empirical strategy;
- Identify and collect appropriate data (e.g., via API or web scraping using Python);
- Clean, structure, and prepare the dataset for analysis;
- Conduct exploratory data analysis, visualization, and regression analysis;
- Present findings and managerial insights in a structured academic paper.

Grading Scale

Grade	Percentage Range
A	≥ 93
A ⁻	90 – 92.9
B ⁺	86 – 89.9
B	83 – 85.9
B ⁻	80 – 82.9
C ⁺	76 – 79.9
C	70 – 75.9
D	60 – 69.9
F	≤ 59.9
IN	Incomplete

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